AdaptiveQKE: Context-Aware Hybrid Key Exchange for TLS Channels

Final Year Project Presentation

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Presentation Outline

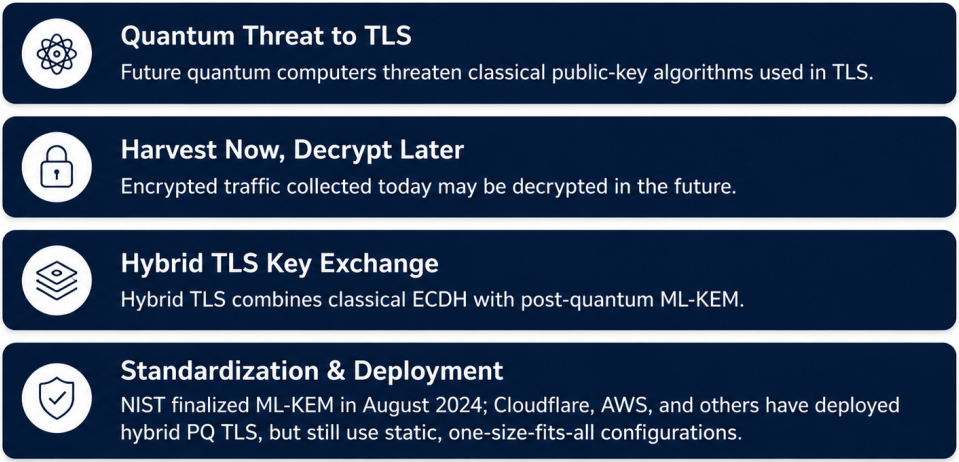
1. INTRODUCTION & MOTIVATION
2. SYSTEM DESIGN & METHODOLOGY
3. EXPERIMENTAL SETUP & PERFORMANCE METRICS
4. RESULTS, FUTURE WORK & CONCLUSION

Introduction & Motivation

From the Quantum Threat to Adaptive Hybrid Group Selection

1. **Background**
2. **Problem Statement**
3. **Proposed Solution**
4. **Project Objectives**

Background



[1] [2] [3]

Problem Statement



Unnecessary handshake cost
Higher latency, CPU time, or handshake size.

Handshake failure risk
May increase under constrained conditions.

Missed stronger security
A stronger group may be possible but unused.

Device capabilities and network conditions differ across environments.

[2]

Proposed Solution

Select the suitable hybrid group before the TLS handshake.



Two-stage policy: RTT-driven candidate, then device-aware validation — no changes to the standard TLS 1.3 handshake.

[4] [5]

Project Objectives

01

Design & Implement the Selection Layer

Policy-driven, pre-handshake hybrid-group selection — no TLS 1.3 modification.



02

Evaluate Adaptive vs. Static Configurations

Compare adaptive selection against static baselines across device and network conditions.

METRICS: **Latency** **CPU time** **Handshake size**

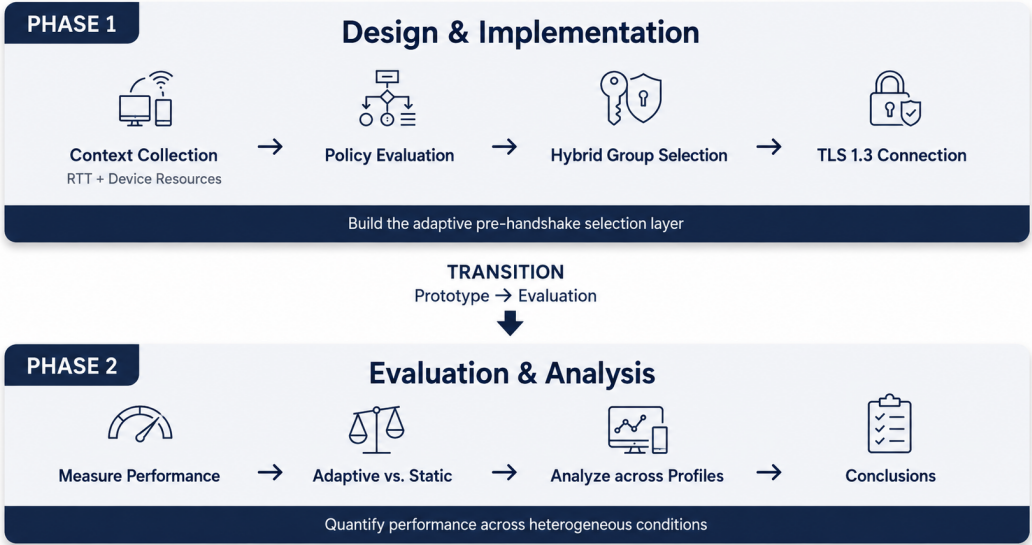


System Design & Methodology

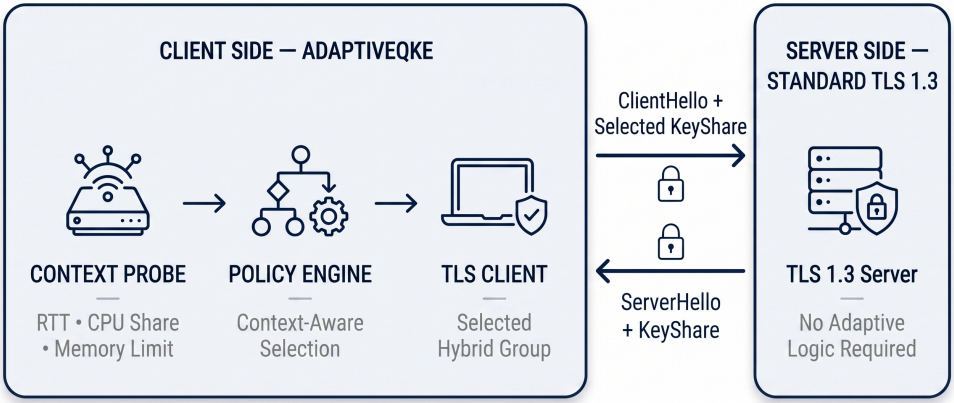
System Architecture, Hybrid Key Exchange,
and Policy-Driven Group Selection

1. **Two-Phase Methodology**
2. **System Architecture**
3. **Hybrid Key Exchange**
4. **Two-Stage Policy Engine**

Two-Phase Methodology

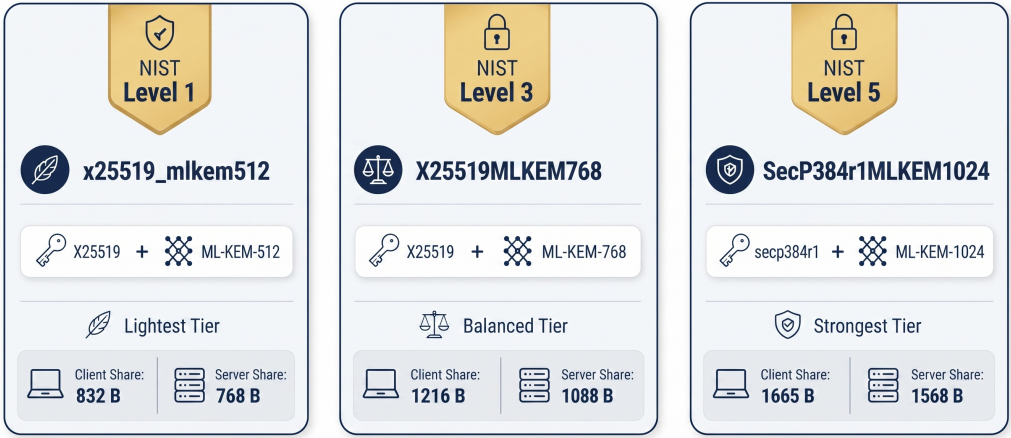


AdaptiveQKE System Architecture



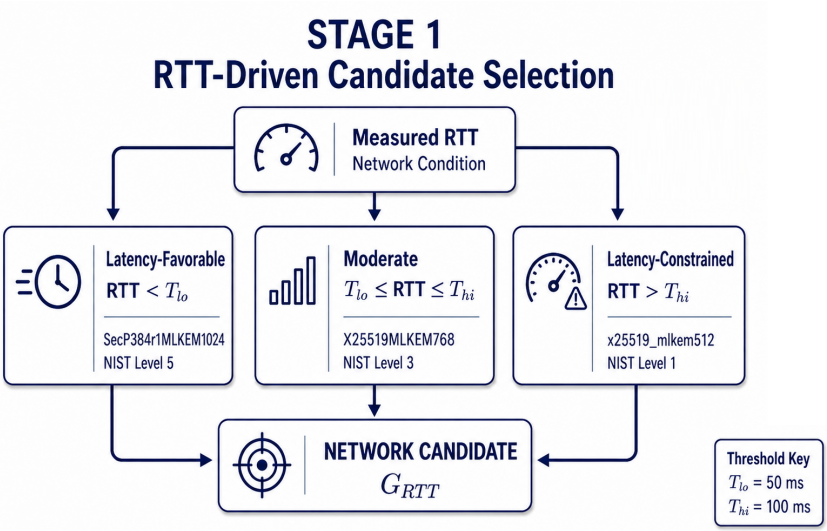
Adaptive intelligence before the handshake - Standard TLS 1.3 handshake after selection

Hybrid Key Exchange Allow-List



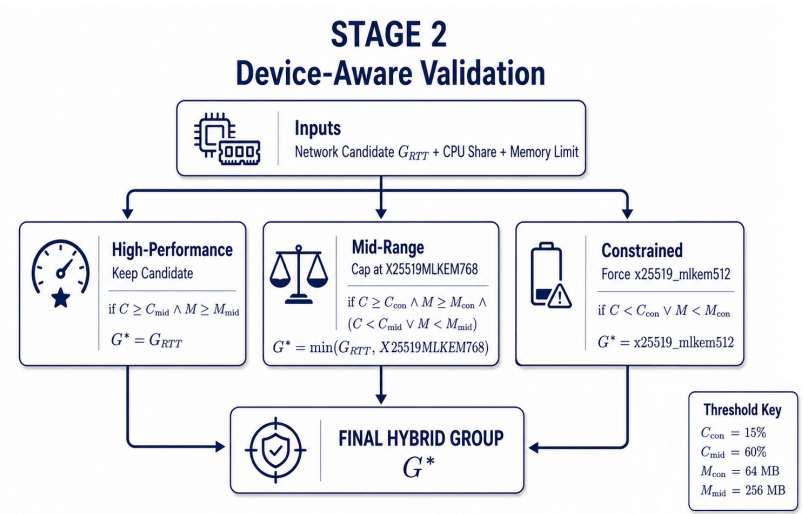
Increasing security tiers introduce higher communication and processing cost.

Policy Engine: Stage 1



RTT determines the initial network candidate before device-aware validation.

Policy Engine: Stage 2



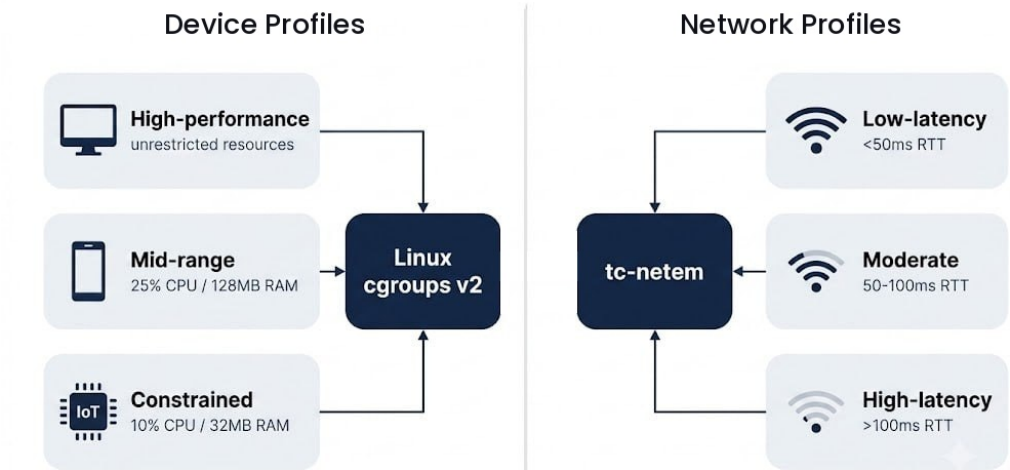
Device capability validates the final hybrid group before the TLS 1.3 handshake.

Experimental Setup & Performance Metrics

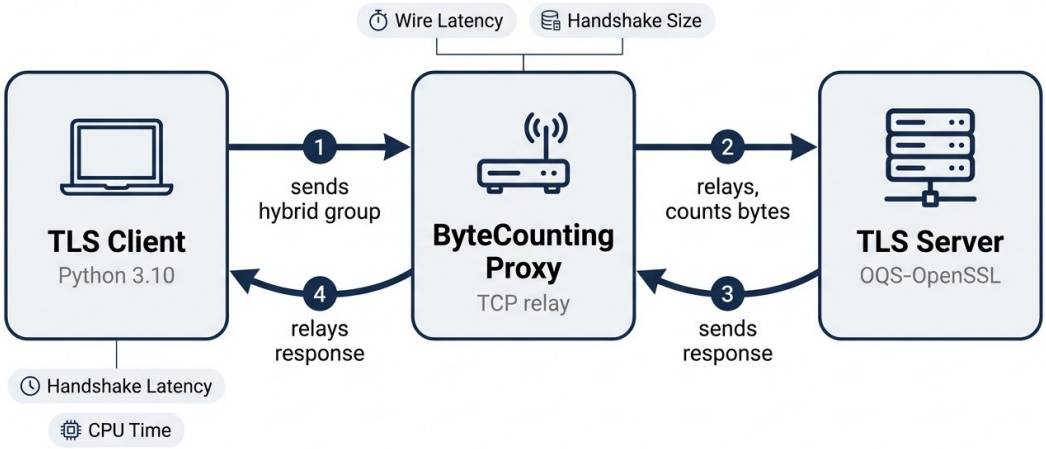
Controlled Device and Network Profiles,
Experimental Workflow, and Evaluation Metrics

1. **Device & Network Profiles**
2. **Testbed Architecture**
3. **Configurations & Scale**
4. **Performance Metrics**

Experimental Setup — Device & Network Profiles



Experimental Testbed Architecture



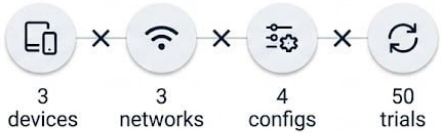
Device profile enforced via Linux cgroups v2 | Network condition enforced via **tc-netem**

Evaluation Configurations & Scale

Configurations Compared

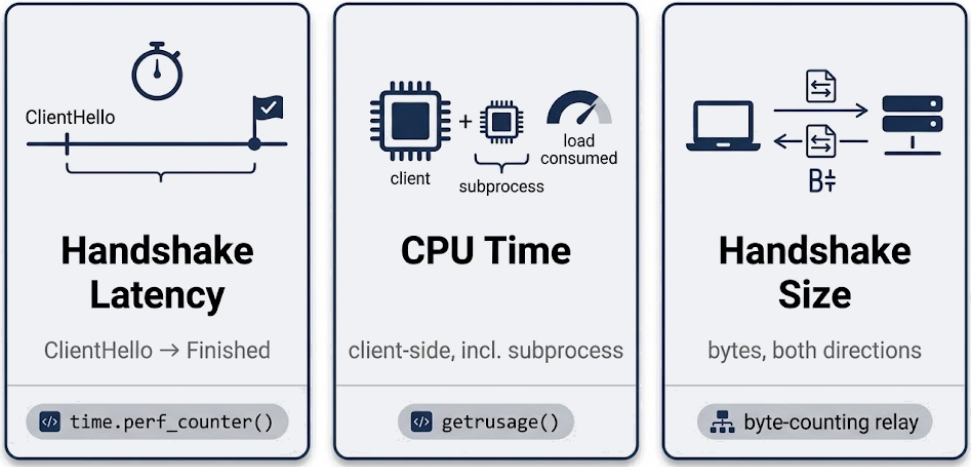
- Static-512
- Static-768
- Static-1024
- **AdaptiveQKE**
  policy-selected per connection

Experiment Scale



= 1,800 handshakes

Performance Metrics

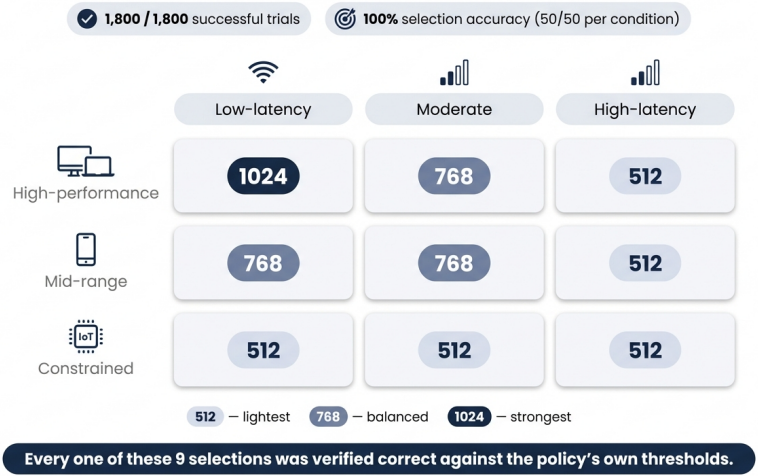


Results, Future Work & Conclusion

Experimental Findings,
AdaptiveQKE Evaluation, Conclusions, and Future Directions

1. **Selection Accuracy & Behavior**
2. **Overhead Relative to Static Baseline**
3. **Limitations & Future Work**
4. **Conclusion**

Selection Accuracy and Behavior



Overhead Relative to Each Static Baseline

vs. Static-512

- ↑ Handshake size: **+13.5%**
- ↔ CPU time: **about even** (+2.1%)
- ↔ Wire latency: **about even** (+2.0%)

vs. Static-768

- ↓ Handshake size: **−7.4%**
- ↔ CPU time: **about even** (+2.7%)
- ↔ Wire latency: **about even** (+0.4%)

vs. Static-1024

- ↓ Handshake size: **−26.2%**
- ↓ CPU time: **−16.9%**
- ↓ Wire latency: **−10.9%**

One active RTT probe per connection (~72ms) --- a production deployment could derive this passively from existing TCP telemetry. The policy decision itself is negligible (~0.03ms).

AdaptiveQKE lands at whichever tier the device and network can support --- not the cheapest option, not the strongest by default.

Limitations & Future Work

	LIMITATIONS	FUTURE WORK
01	No application-defined security bounds (Stage 0) — selection is performance-driven only	Add Stage 0: security bounds (min/max)
02	Network paths and device tiers are emulated, not physical hardware	Validate on real devices and networks
03	Decision thresholds were set by the same team evaluating them	Data-driven threshold tuning
04	No formal security analysis performed	Formal security analysis

Conclusion

AdaptiveQKE

Context-aware hybrid-group selection
before the TLS 1.3 handshake



01 - Selection Accuracy

100% correct across every tested condition



02 - Matching Static Behavior

No measurable latency added after selection



03 - Standard TLS 1.3 Flow Preserved

No changes required to the TLS 1.3 handshake

Across all nine tested device-network conditions, AdaptiveQKE selected the configuration that best balanced security level against handshake cost for the conditions at hand.

References

1. National Institute of Standards and Technology
Module-Lattice-Based Key-Encapsulation Mechanism Standard
FIPS 203, 2024

2. Amazon Web Services
Using Hybrid Post-Quantum TLS with AWS KMS
Developer Guide, 2024

3. Cloudflare, Inc.
Post-Quantum Cryptography — SSL/TLS Documentation
2024

4. Westerbaan, B. et al.
Post-Quantum Hybrid ECDHE-MLKEM Key Agreement for TLSv1.3
draft-ietf-tls-ecdhe-mlkem-04, 2025

5. Rescorla, E.
The Transport Layer Security (TLS) Protocol Version 1.3
RFC 8446, 2018

6. Open Quantum Safe
ML-KEM Algorithm Details
2024

Thank You!

Questions & Answers